Reviewer 1 - Initial submission, May 25

I cannot recommend this manuscript for publication. As detailed in my major comments, the core assumptions of the proposed framework conflict with established principles of general relativity , the mass-reconstruction method violates basic mass conservation, and the \zeta–mass relation is neither theoretically motivated nor uniquely determined. The interpretation of the BTFR normalization is inconsistent with observations, and several implications in Section 9 contradict the model's own assumptions. Taken together, these issues indicate that the framework is not internally coherent and does not provide a physically viable alternative to existing theories.

My major Comments

1. Introduction

1.1

The manuscript proposes that the observed geometry is shaped by "accumulated metric changes" along a photon's path. This idea is intuitively appealing, but it is fundamentally at odds with the structure of general relativity. In GR, spacetime curvature is determined locally by the energy–momentum tensor, and photons simply follow null geodesics defined by that curvature. They do not carry information about how the metric "evolves" along their trajectory, nor do they integrate or transport metric distortions from one region to another. I am not aware of any formulation in either GR or mainstream modified gravity theories where the observed geometry is generated by photons "accumulating" metric evolution along their paths. Line-of-sight integrals of the potential or curvature certainly appear in gravitational lensing and CMB physics, but there the metric is determined by the field equations, and photons merely probe it by following null geodesics. The mechanism proposed here inverts this logic: it effectively promotes photons from probes of geometry to carriers of metric evolution, which does not fit into the usual geometric framework of GR or its common extensions.

1.2 I understand why the authors are drawn to the Painlevé–Gullstrand (PG) form: it offers a very clear and intuitive picture of free fall in a spherically symmetric spacetime. As a simplified model, it can certainly provide useful intuition. At the same time, it is worth noting that the PG metric relies on rather specific assumptions—namely a static, spherically symmetric vacuum exterior. Real disk galaxies, with their extended mass distributions and significant rotation, depart from these conditions in important ways.

This does not mean the PG picture is without value, but it does mean that its direct application to galactic dynamics requires careful justification. A brief discussion acknowledging the limitations of this approximation, and clarifying which aspects of the PG intuition the authors intend to retain, would help readers better understand how the model is meant to be used and where its applicability naturally ends.

2. Section 3

2.1 The manuscript repeatedly treats the PG "flow" as if it were a physical motion of space. While the intuitive appeal of this picture is understandable, the analogy is taken too far and ends up assigning physical meaning to what is, in GR, purely a coordinate construction. In the PG slicing, v(r)= sqrt(2GM/r),is simply the radial free-fall velocity of a test particle released from rest at infinity. Reinterpreting this quantity as “the free-fall speed of the metric itself” or as “the rate at which space streams toward the gravitational potential” has no support in GR. The metric does not fall, and space does not flow like a fluid; the PG “river” is a pedagogical device, not a physical medium. Moreover, the PG free-fall velocity describes radial infall, whereas the phenomena the authors aim to explain—galactic dynamics—are governed by tangential, stable circular motion. Treating the PG free-fall speed as dynamically relevant on galactic scales conflates coordinate behavior with actual gravitational motion.

The manuscript further relies on two assumptions associated with the so-called “Mezzi effect,” introduced by the authors in a separate work. The first assumption—that light carries geometric distortion information accumulated along the entire line of sight—is unconventional and difficult to reconcile with the locality and geodesic structure of general relativity. In GR, photons follow null geodesics determined by the local metric; they do not integrate or transport metric evolution along their paths. Even though similar statements appear in a small number of recent papers, repetition in the literature does not resolve the underlying inconsistency with established geometric principles.

The second assumption introduces a “vacuum compliance” parameter intended to convert flow-induced stress into observable geometric distortion. While this is presented as a phenomenological extension, its physical motivation and mathematical definition remain insufficiently developed. It is unclear how this compliance relates to known response functions or effective descriptions, and without such grounding the construction appears ad hoc.

Taken together, these assumptions form the core of the proposed mechanism, yet both currently lack the theoretical justification required for broader acceptance. Substantial clarification and a more rigorous connection to established principles would be necessary before the Mezzi effect can be meaningfully assessed or incorporated into the present argument.

2.2 The manuscript introduces the concept of “vacuum compliance” as a new physical ingredient. While proposing novel parameters or phenomenological extensions is not inherently problematic, the physical status of this quantity remains unclear. In its current form, the compliance, C, is introduced as an intrinsic property of the vacuum, yet no theoretical motivation is provided from general relativity, quantum field theory, or any established modified-gravity framework. As a result, it is difficult to assess whether C represents a meaningful physical scale or simply functions as a model-dependent fitting parameter calibrated to the SPARC sample.

This point is not intended to discourage creative theoretical exploration. Rather, for such an idea to gain broader traction, the manuscript would benefit from a clearer discussion of the physical assumptions underlying the notion of vacuum compliance, its possible theoretical origin, and whether it can be constrained or tested independently of the rotation-curve fitting. Without such clarification, the interpretation of C=377 as anything beyond an empirical parameter remains uncertain.

3. Section 4

3.1 The most serious problem in the mass–reconstruction procedure is that it violates mass conservation in a fundamental way. The method assumes that the observed surface density, Σ_{obs}, represents the true physical density, while the coordinate grid is compressed by the Mezzi effect. When the “true” area of a shell is expanded by a factor $R_j>1$, the authors simply multiply the observed mass element by this geometric factor: $\Delta M_{true,j} = \Delta M_{obs, j}R_j$. This implies that the physical mass of a shell increases solely because the coordinate area is rescaled.In other words, the procedure effectively creates mass through a coordinate transformation. This is incompatible with any physically meaningful notion of mass conservation or with the covariant conservation law

$$\nabla_\mu T^{\mu\nu} =0$$

in general relativity. A coordinate rescaling cannot generate additional mass, and surface density cannot remain invariant under a non-trivial geometric distortion. As a result, the reconstructed “true” mass profile is not physically meaningful, and the procedure cannot be interpreted as a valid mass–recovery method.

3.2 A further conceptual problem arises from the assumption that the observed surface density, Σ_{obs}, represents the true physical mass density of a shell. Surface density is not a geometric invariant; it is a derived observational quantity that depends on photon flux, projected area, distance calibration, line-of-sight integration, and instrumental response. If the spatial metric is distorted—as the manuscript proposes—then the projected area, photon arrival rate, and inferred luminosity density must all change accordingly. Therefore, Σ_{obs} cannot remain unchanged under a non-Euclidean geometric transformation.

Treating Σ_{obs} as a fixed physical density while simultaneously expanding the true area leads directly to the unphysical conclusion that the mass increases by a factor R_j. This is the violation of mass conservation I mentioned above. The assumption that an observationally defined quantity behaves as a coordinate-independent physical density is ad hoc and incompatible with basic principles of radiative transfer and mass conservation.

3.3 A major conceptual issue concerns the origin and uniqueness of the relationship between the Mezzi scale factor $\zeta(r)$ and the baryonic mass of galaxies. This relationship is not theoretically derived but is first implicitly imposed in Section 4, where $\zeta(r)$ is used to rescale radii, areas, and ultimately the enclosed mass. From this point onward, all key results—mass amplification, deflection angles, the RAR mapping, and the rotation-curve reconstruction—depend critically on the assumed ζ–mass connection.

However, the ζ–mass relation exhibits very large scatter in the sample, indicating that it is not uniquely determined by the data. The choice of the constant $C=377$, which sets the overall amplitude of $\zeta(r)$, appears to be selected to optimize agreement with the SPARC sample rather than derived from first principles. Different choices of C would produce different ζ, different mass-amplification factors, and therefore different predictions for lensing, the RAR, and rotation curves.

Because the \zeta–mass relation is neither theoretically constrained nor uniquely determined by the observations, the framework currently lacks predictive power. The manuscript should explicitly justify how \zeta(r) and the constant C are chosen, explain why this choice is preferred over other possible mappings, and demonstrate that the results do not depend sensitively on this tuning. Without such clarification, it is difficult to assess whether the Mezzi effect represents a physical mechanism or an empirical rescaling calibrated to match the data.

4. Section 5

4.1 A central concern arises in the treatment of gravitational deflection. The manuscript appears to conflate the role of the three-dimensional galactic scale with the quantity that actually governs light bending in general relativity. In the weak-field limit, the deflection angle depends on the projected mass enclosed within the impact parameter, not on the physical size of the galaxy. The impact parameter is simply the closest approach of the light ray, and enlarging the three-dimensional radius of the system does not alter this quantity. As long as the projected mass increases, the deflection angle must increase accordingly.

In the proposed framework, however, the authors argue that the enhancement in mass can be offset by an increase in the “true’’ galactic radius, leading in some cases to little or no change in the predicted deflection angle. This interpretation is difficult to reconcile with the standard GR definition of weak-lensing deflection, where the three-dimensional extent of the mass distribution plays no direct role. If the true mass is indeed several times larger than the observed mass, the corresponding lensing signal should be stronger, independent of the system’s intrinsic spatial scale.

Given this discrepancy, the manuscript should clearly explain how the Mezzi framework modifies the GR lensing relation, why the usual dependence on projected mass no longer applies, and how the proposed behavior remains consistent with existing weak-lensing observations. Without such clarification, it is challenging for the reader to understand why the new result should be preferred over the well-tested GR prediction.

5. Section 8

5.1 A major concern arises from the interpretation of Figures 20-21. While the authors emphasize that the \beta-\zeta relation in the true frame exhibits an approximately zero slope, the vertical scatter of the normalization offset is extremely large, spanning roughly 1–3 dex. This level of dispersion is physically implausible for the baryonic Tully–Fisher relation. Observationally, the BTFR is one of the tightest empirical relations in galaxy dynamics, with a typical intrinsic scatter of only 0.1–0.2 dex (Lelli et al. 2019 MNRAS). Even the most extreme outliers rarely exceed 0.3 dex.

A \beta offset varying by 1–3 dex corresponds to mass discrepancies of one to three orders of magnitude. Such a wide spread indicates that the reconstructed “true” masses are highly unstable and that the \zeta–mass mapping lacks uniqueness or physical constraint. In this context, the disappearance of the slope in Figures 20-21 cannot be taken as evidence that the

BTFR normalization has been "restored." Instead, the enormous scatter demonstrates that the model fails to reproduce the defining tightness of the BTFR, which is essential to any credible explanation of its physical origin.

In short, the conclusion drawn from Figures 20-21 is not justified. A zero slope with a scatter of several dex does not support the proposed geometric interpretation; rather, it highlights a fundamental inconsistency in the mass reconstruction and the assumed \zeta(r) scaling.

6. Section 9
The implications in Section 9 are not internally consistent with the assumptions of the Mezzi framework. The main contradictions are:
6.1 The model assumes a spherically symmetric vacuum flow, yet the inclination-dependent dark matter fraction requires a strongly anisotropic, disk-plane–dominated compression. These two assumptions cannot both be true.
6.2 Earlier sections state that\zeta(r) is determined solely by the galaxy's own baryonic potential. Section 9.2 then claims \zeta(r) is strongly modified by the host cluster potential. This contradicts the model's own definition of \zeta(r).
6.3 The model states that higher concentration produces stronger compression (smaller \zeta). Section 9.4 claims compact globular clusters should have weak compression (\zeta approximately 1). This reverses the model's own scaling rule.
6.4 The mathematical form of \zeta(r) steepens the inner mass profile (enhanced compression). Section 9.5 claims it naturally produces cores. The predicted geometry actually produces cusps, not cores.
6.5 The model asserts \zeta(r) only rescales coordinates without altering physical dynamics. Section 9.6 requires \zeta(r) to change the true mass distribution enough to fit Einstein radii. A pure coordinate transformation cannot generate additional lensing mass.

Dear Editor

Thank you for considering our manuscript.

The reviewer evaluates the model against the strict theoretical foundations of General Relativity, but our framework is explicitly presented as phenomenological. The reviewer entirely ignores the primary result: that the model reproduces galaxy dynamics and uncovered clean mathematicalscaling relations, with a single free parameter.

Only two comments are made on the papers results, 1. On lensing: They argue we offset mass with an "increased radius." This is incorrect; we propose a geometric correction, not a physical radius increase. 2. On BTFR offset slope: They claim the scatter is high, yet scatter is consistent with the observed SPARC data plot in the paper and the rest of the results presented in the paper.

Because the review attacks claims not made in the paper and ignores the empirical success of the results, I respectfully request the opportunity to reply to the reviewer.

Sincerely,
Dr. Brahim Benaissa

Manuscript Number: **DARK-D-26-00588**

On the Observer-Frame Interpretation of Radial Coordinates in Gravitational Potentials

Dear Author,

In view of your request to have a chance to reply the referee regarding referee's criticism and recommendation of rejection, we decide to give you a chance to reply the Report and revise your paper.

Please resubmit your revised manuscript by **Jun 27, 2026**.

Physics of the Dark Universe values your contribution and I look forward to receiving your revised manuscript.

Kind regards,
Editor
Physics of the Dark Universe

Reviewer
I cannot recommend this manuscript for publication. As detailed in my major comments, the core assumptions of the proposed framework conflict with established principles of general relativity, the mass-reconstruction method violates basic mass conservation, and the \zeta–mass relation is neither theoretically motivated nor uniquely determined. The interpretation of the BTFR normalization is inconsistent with observations, and several implications in Section 9 contradict the model's own assumptions. Taken together, these issues indicate that the framework is not internally coherent and does not provide a physically viable alternative to existing theories.

Author
While the critique raises fundamental questions, it does not address the framework's phenomenological foundation. Several of the core objections arise from misunderstandings of the model's mechanics: such as, conflating the metric flow velocity with the orbital velocity of the stars, equating the BTFR normalization offset with BTFR scatter, and using the model's stated boundary conditions (spherical and isolated) to contradict the theoritical implications. Most importantly, the "mass creation" objection misinterprets local density conservation as the generation of new photons. We spent particular effort on this point in the revised manuscript. Below we provide point by point responses, accompanied by corresponding manuscript revisions.

1. Introduction

Q 1.1a
The manuscript proposes that the observed geometry is shaped by "accumulated metric changes" along a photon's path. This idea is intuitively appealing, but it is fundamentally at odds with the structure of general relativity. In GR, spacetime curvature is determined locally by the energy–momentum tensor, and photons simply follow null geodesics defined by that curvature. They do not carry information about how the metric "evolves" along their trajectory, nor do they integrate or transport metric distortions from one region to another.

A 1.1a.
The proposed framework does not modify Einstein's field equations. Rather, we suggest that the measurement process, the mapping between source frame and observed frame, is path dependent. This is conceptually analogous to gravitational lensing, the deflection angle depends on \int \nabla_\perp \Phi \, dl, a line of sight integral that is not interpreted as the photon "carrying" gravitational potential.

We add the following paragraph to Section 3.2, after Eq. 7:

"It is important to distinguish this accumulation from a dynamical non locality. The integral A(r) is not a causal influence transported by the photon; it is a theoretical path dependent measurement, analogous to the optical path length in gravitational lensing. The null geodesic equation k^\mu \nabla_\mu k^\nu = 0 remains locally valid at every point along the trajectory. What the Mezzi effect modifies is the mapping between the source frame and the observer frame, not the local dynamics."

Q 1.1b
I am not aware of any formulation in either GR or mainstream modified gravity theories where the observed geometry is generated by photons "accumulating" metric evolution along their paths. Line-of-sight integrals of the potential or curvature certainly appear in gravitational lensing and CMB physics, but there the metric is determined by the field equations, and photons merely probe it by following null geodesics. The mechanism proposed here inverts this logic: it effectively promotes photons from probes of geometry to carriers of metric evolution, which does not fit into the usual geometric framework of GR or its common extensions.

A 1.1b
This is precisely its status as a phenomenological proposal. The scale factor \zeta(r) is not a coordinate transformation in the GR sense. The scale factor describes a “measurement distortion”. The physical process of mapping electromagnetic signals to source positions is biased by the proposed vacuum compliance. The photon remains a probe. The galaxy itself is unchanged. Its observed image is distorted. This work explores whether such a distortion, applied consistently across all observables (photometry, kinematics, lensing), can explain the apparent mass discrepancy without invoking additional matter or modified dynamics.

Q 1.2a
I understand why the authors are drawn to the Painlevé–Gullstrand (PG) form: it offers a very clear and intuitive picture of free fall in a spherically symmetric spacetime. As a simplified model, it can certainly provide useful intuition. At the same time, it is worth noting that the PG metric relies on rather specific assumptions, namely a static, spherically symmetric vacuum exterior. Real disk galaxies, with their extended mass distributions and significant rotation, depart from these conditions in important ways.

A 1.2a
The PG velocity field v(r) = -\sqrt{2GM(r)/r} calculates the zero order radial gradient of the gravitational potential. The spherical approximations are routinely applied to disk galaxies for mass modeling, dynamical estimates, and dark matter halo fitting. The critical test in this work is empirical. i.e. if the spherical PG flow were fundamentally inappropriate for disk galaxies, then

applying a single, globally fixed compliance parameter C across 175 galaxies with highly diversity (stellar masses 10^7–10^{11} $M_\odot$, velocities 20–300 km/s, surface brightness spanning three orders of magnitude) would produce catastrophic failure. This did not happen, instead the proposed framework produces a tight ζ-mass correlation ($r=0.96$), recovers the RAR, and transforms the BTFR slope from ~2.8 to ~3.7.

The intermediate and high mass galaxies exhibit increased scatter and systematic bias (Section 6.1), which we attribute precisely to the spherical symmetry and isolation assumptions. These deviations are not failures of the framework; they will be addressed in the next order models.

We expand Section 3.5 with the following paragraph:

"The spherical symmetry assumption is a zero order approximation. This work tests whether a spherical model can capture the geometric distortion. An axisymmetric extension, replacing $v(r)$ with a flattened potential, is the natural next step."

Q 1.2b
This does not mean the PG picture is without value, but it does mean that its direct application to galactic dynamics requires careful justification. A brief discussion acknowledging the limitations of this approximation, and clarifying which aspects of the PG intuition the authors intend to retain, would help readers better understand how the model is meant to be used and where its applicability naturally ends.

A 1.2b
Section 3.5, explicitly titled “Methodological Assumptions and Boundary Conditions”, already addresses this limitation the reviewer asks for, specifically listing "Spherical Symmetry" and "Isolation of the Source" as critical simplifying assumptions that bound the model's domain of applicability.

The framework does not claim that real galaxies are spherically symmetric; it models, at this stage, the “geometric distortion” by a spherical model.

Q 2.1a
The manuscript repeatedly treats the PG "flow" as if it were a physical motion of space. While the intuitive appeal of this picture is understandable, the analogy is taken too far and ends up assigning physical meaning to what is, in GR, purely a coordinate construction. In the PG slicing, $v(r) = \sqrt{2GM/r}$, is simply the radial free-fall velocity of a test particle released from rest at infinity. Reinterpreting this quantity as "the free-fall speed of the metric itself" or as "the rate at which space streams toward the gravitational potential" has no support in GR. The metric does not fall, and space does not flow like a fluid; the PG "river" is a pedagogical device, not a physical medium.

A 2.1a

In standard GR, the PG velocity is a coordinate artifact. In this framework however, the vacuum possesses a compliance to the kinematic stress encoded by $v(r)$. This is not a derivation from GR; it is a “new physical hypothesis”. As detailed in Section 3.4, we discuss that the vacuum behaves as a compliant medium with rheological memory, where the spatial dependence of the velocity field encodes convective inertia. The validity of this work is judged by its predictive power, i.e. a single parameter C reproduces rotation curves, scaling relations, and lensing corrections across 175 galaxies. This work is explicit about its phenomenological status; it does not claim to derive vacuum compliance from Einstein's equations.

We add to Section 3.4:

"We emphasize that this is a phenomenological reinterpretation, not a derivation from GR. In standard GR, the PG velocity $v(r)$ is the shift vector in a spatially flat coordinate system. The Mezzi framework hypothesizes that this kinematic field encodes a real physical property of the vacuum, its compliance to geometric stress. This hypothesis generates testable predictions in this paper, that are not imposed but emerge from the mathematics"

Q 2.1b

Moreover, the PG free-fall velocity describes radial infall, whereas the phenomena the authors aim to explain, galactic dynamics, are governed by tangential, stable circular motion. Treating the PG free-fall speed as dynamically relevant on galactic scales conflates coordinate behavior with actual gravitational motion.

A 2.1b

This objection conflates the velocity of the metric flow with the orbital velocity, which move tangentially within that geometry. Section 3.1 explicitly defines the velocity field as "the free fall speed of the “metric itself”, which is used to compute the Mezzi scale factor ζ. Section 6 clearly separates this from stellar dynamics by assuming the Validity of Newtonian Dynamics, where orbital velocities obey the Keplerian relation (tangential motion).

In standard physics, it is precisely the presence of a radial inward pull that sustains stable tangential circular motion. The radial gravitational force GM/r^2 sustains tangential circular motion via the centripetal condition $v_{\text{orb}}^2/r = GM/r^2$. The PG velocity measures the potential depth that generates this force; it does not drive the tangential motion.

Q2.1c

The manuscript further relies on two assumptions associated with the so-called "Mezzi effect," introduced by the authors in a separate work. The first assumption, that light carries geometric distortion information accumulated along the entire line of sight, is unconventional and difficult to reconcile with the locality and geodesic structure of general relativity. In GR, photons follow null geodesics determined by the local metric; they do not integrate or transport metric evolution along their paths. Even though similar statements appear in a small number of recent papers, repetition in the literature does not resolve the underlying inconsistency with established geometric principles.

A 2.1c
The reviewer notes that light carrying geometric distortion is difficult to reconcile with the strict locality of standard GR. The framework is presented as a phenomenological exploration, not a theorem within GR. The accumulation integral $A(r)$ is not a causal influence transported by the photon; it is a path dependence measurement effect. In standard gravitational lensing, the deflection angle depends on $\int \nabla_\perp \Phi \, dl$, a line-of-sight integral of the gravitational potential. No one interprets this as the photon "carrying" potential information; rather, the photon's trajectory samples the potential along its path, and the observer's inference of lens mass is path dependent. The Mezzi effect operates analogously: the observer's inference of source geometry is path dependent. The photon does not modify the metric behind it; the null geodesic equation $k^\mu \nabla_\mu k^\nu = 0$ remains locally valid. What is suggested is a frame transformation between source and observer, which acquires path dependence through $A(r)$.

We add the lensing analogy explicitly to Section 3.2:

"The accumulation integral $A(r)$ is analogous to the lensing potential $\psi(\boldsymbol{\theta}) = \int \nabla_\perp \Phi \, dl$ in standard gravitational lensing. In both cases, the photon follows local null geodesics; the observer's inference, of lens mass or source geometry, depends on the integrated environment along the trajectory. The Mezzi effect extends this path dependence from the deflection angle to the coordinate scale itself."

Q2.1d
The second assumption introduces a "vacuum compliance" parameter intended to convert flow-induced stress into observable geometric distortion. While this is presented as a phenomenological extension, its physical motivation and mathematical definition remain insufficiently developed. It is unclear how this compliance relates to known response functions or effective descriptions, and without such grounding the construction appears ad hoc.

A2.1d
The physical motivation is explicitly phenomenological; the framework introduces this extension to capture empirical results, as is standard for new scaling parameters. The rheological analogy in Section 3.4 provides an analogy for how a medium with memory can respond to accumulated stress.

The Mezzi framework relate to temporal history with spatial path history: the stress $\sigma(r) = K(r)-1$ accumulates along the photon trajectory from source to observer, and the compliance C converts this history dependent stress into observable strain $\varepsilon(r) = \ln \zeta(r)$.

Q2.2a
The manuscript introduces the concept of "vacuum compliance" as a new physical ingredient. While proposing novel parameters or phenomenological extensions is not inherently problematic, the physical status of this quantity remains unclear. In its current form, the compliance, C, is introduced as an intrinsic property of the vacuum, yet no theoretical motivation is provided from general relativity, quantum field theory, or any established modified-gravity framework.

A2.2a
C is not derived from GR, QFT, or any modified gravity. Its status “phenomenological”, analogous to numerous fundamental constants that emerged from empirical models before theoretical derivation. The Hubble constant H_0 was a fitted parameter before cosmic expansion; the cosmological constant Λ was introduced phenomenologically before dark energy; MOND's a_0 remains without consensus derivation.

The Mezzi framework proposes C as a candidate for similar status: a universal property of the vacuum determined by its response to kinematic stress. Theoretical derivation might be identified in future work.

We add to Section 3.6:

"The compliance C is presented as a phenomenological constant, analogous to a_0 in MOND or Λ in ΛCDM. Its theoretical derivation is not discussed this work "

Q2.2b
As a result, it is difficult to assess whether C represents a meaningful physical scale or simply functions as a model-dependent fitting parameter calibrated to the SPARC sample.

A2.2b
A fitting parameter is adjusted per system to force agreement; C is rigidly fixed across the entire sample. The SPARC database spans four orders of magnitude in stellar mass, three orders of magnitude in surface brightness, and velocities from 20 to 300 km/s.
Standard ΛCDM requires 2–3 free parameters per galaxy (halo mass, concentration, core radius). The proposed framework reproduces this diversity with “one universal constant”. The mathematical structure of the path integral (Eqs. 5–7) and compliance scaling (Eq. 12) produce a tight ζ-mass correlation (r=0.96) and the emergence of clean mathematical scaling relations.

Q2.2c

This point is not intended to discourage creative theoretical exploration. Rather, for such an idea to gain broader traction, the manuscript would benefit from a clearer discussion of the physical assumptions underlying the notion of vacuum compliance, its possible theoretical origin, and whether it can be constrained or tested independently of the rotation-curve fitting. Without such clarification, the interpretation of C=377 as anything beyond an empirical parameter remains uncertain.

A2.2c

The paper does not claim C is derived from existing theory; it claims it is a single parameter that, once fixed, reproduces multiple independent observables. This objection conflates theoretical derivation with phenomenological discovery. The manuscript identifies (In section 8.2) four simultaneous constraints used to determine C , namely a vanishing true-frame BTFR normalization slope, minimized rotation curve residuals, lensing slope ≈ 2 relative to velocities, and BTFR slope ≈ 4.

Q3.1

The method assumes that the observed surface density, Σ_{obs}, represents the true physical density, while the coordinate grid is compressed by the Mezzi effect. When the "true" area of a shell is expanded by a factor $R_j > 1$, the authors simply multiply the observed mass element by this geometric factor: $\Delta M_{\text{true},j} = \Delta M_{\text{obs},j} R_j$. This implies that the physical mass of a shell increases solely because the coordinate area is rescaled. In other words, the procedure effectively creates mass through a coordinate transformation. This is incompatible with any physically meaningful notion of mass conservation or with the covariant conservation law $\nabla_\mu T^{\mu\nu} = 0$ in general relativity. A coordinate rescaling cannot generate additional mass... As a result, the reconstructed "true" mass profile is not physically meaningful, and the procedure cannot be interpreted as a valid mass-recovery method.

A3.1

This objection does not engage with the framework's core hypothesis. We clarify that the paper propose two frames:

The True Frame: The physical reality where Newtonian dynamics holds and the galaxy extends to larger true radii and contains more mass than inferred in the observed frame .

The Observer Frame: A compressed projection of the true frame caused by the compliant vacuum.

The paper states in Section 4.2: "We posit that the observed surface density Σobs provides the correct mass density for the local shell, but the coordinate grid is compressed, mapping the true extent onto smaller observed radii."

We explicitly distinguish its framework from a mere mathematical coordinate transformation. In Section 3.5, "It is not a reversal of physical compression." In Section 3.4, ζ is defined as a "measurement transfer function that rescales inferred distances based on the integrated propagation history".

We propose a phenomenological effect where the observed mass is an underestimate of the true physical mass due to a "compressed projection". This is not a violation of $\nabla_\mu T^{\mu\nu} = 0$. The local energy momentum tensor is conserved at every point. The discrepancy arises because M_{obs} is an undercount of the baryons due to integrating over a compressed coordinate grid, and M_{true} represents the actual baryonic mass.

To explicitly address the persistent objection, we have added Section 4.3.4 , "The Absence of a Mass Creation Mechanism". This new section establishes three rigorous no go arguments grounded in the approximate surface brightness invariance of the weak-field regime:

1. No go from approximate Liouville behavior: Surface brightness invariance is not a postulate; it is an approximate consequence of GR geometric optics for $v/c \ll 1$. If the Mezzi effect created mass, it would necessarily alter the photon geodesic bundle or introduce scattering, violating the specific intensity conservation that we explicitly rely on.

2. No go from dimensional analysis: The compliance C and scale factor $\zeta(r)$ are dimensionless. Mass creation would require a source term in the field equations with dimensions of energy density $[T_{00}]$. No such term appears in the Mezzi formalism; the vacuum flow velocity $v(r)$ is purely kinematic and does not contribute to $T_{\mu\nu}$ as an independent energy-momentum source.

3. No go from the equivalence principle: Mass creation would violate the weak equivalence principle. The Mezzi effect preserves it: photons and massive particles traverse the same flowing vacuum, and both the deflection angle α_{true} and circular velocity v_{true} are consistent with the same M_{true}.

Q3.2

A further conceptual problem arises from the assumption that the observed surface density, Σ_{obs}, represents the true physical mass density of a shell. Surface density is not a geometric invariant; it is a derived observational quantity that depends on photon flux, projected area, distance calibration, line-of-sight integration, and instrumental response. If the spatial metric is distorted, as the manuscript proposes, then the projected area, photon arrival rate, and inferred luminosity density must all change accordingly. Therefore, Σ_{obs} cannot remain unchanged under a non-Euclidean geometric transformation. Treating Σ_{obs} as a fixed physical density while simultaneously expanding the true area leads directly to the unphysical conclusion that the mass increases by a factor R_j. This is the violation of mass conservation I mentioned above. The assumption that an observationally defined quantity behaves as a coordinate-independent physical density is ad hoc and incompatible with basic principles of radiative transfer and mass conservation.

A3.2

In Section 4.2 we assume that the observed surface density Σ_{obs} provides the correct mass density for the local shell." The photons received from a given shell carry the local surface density information, but their radial attribution is distorted.

The claim of this work is that, when standard analysis integrates mass using the compressed radial scale, it integrates the correct local density over an incorrect volume, producing $M_{\text{obs}} < M_{\text{true}}$. Thus, the "missing mass" is baryonic mass whose spatial distribution was compressed into an artificially small observed volume.

The mechanism is as follows. The observed surface brightness $I_{\text{obs}}(\theta)$ at angular position θ is related to the true emissivity $j_{\text{true}}(r)$ by a line of sight integral. Standard analysis assumes Euclidean mapping: $\theta = r/D$.

In the Mezzi framework, this mapping is compressed: $\theta_{\text{obs}}(r_{\text{true}}) = r_{\text{true}} \cdot \zeta(r_{\text{true}})/D$. Because $\zeta < 1$ and is strongest at the center, flux from larger true radii is systematically misattributed to smaller observed radii. The total observed flux is conserved (same photons), but its radial distribution is distorted.

(Sadeghi 2026) support our claim, by proving that any apparent concentration is bounded by étendue conservation. Even in transformation optical media employing extreme coordinate compression, radiance, the optical phase space density of power, cannot increase, as a consequence of Liouville's theorem and phase space volume conservation (https://www.nature.com/articles/s41598-026-42509-9).

To formalize this rgument, we have extended Section 4.2, showing that the reconstruction is a measure-preserving remapping of a fixed baryonic density field:

1. Approximate SBI: For the resolved, optically thin SPARC disks at $3.6\,\mu\text{m}$, $I_{\text{obs}}(r_{\text{obs}}) \approx I_{\text{true}}(r_{\text{true}})$. Because \Upsilon_ is a local property, $\Sigma_{\text{obs}}(r_{\text{obs}}) \approx \Sigma_{\text{true}}(r_{\text{true}})$.

2. Mass Conservation Lemma: $\mathcal{M}_{\text{true}} \approx \int \Sigma_{\text{obs}}(r_{\text{obs}}) \, J(r_{\text{obs}}) \, dA_{\text{obs}}$, where $J(r)$ is the Jacobian determinant. The discrete shell algorithm (Eqs 16-19) is the numerical quadrature of this integral. The mass recovery is strictly the integral of the same physical density over a corrected geometric measure.

3. Stress-Energy Consistency: The framework preserves $\nabla_\mu T^{\mu\nu}_{\text{(true)}} = 0$ exactly in the true frame. The observational mass deficit arises because integrating the approximately invariant density profile over the compressed observational grid truncates the integral prematurely.

3. The Absence of a Mass Creation Mechanism:

- No-go from approximate Liouville behavior: Passive coordinate compression cannot increase phase-space density; photon conservation forbids brightening without sources.
- No-go from dimensional analysis: C and ζ are dimensionless, no energy scale exists to create mass.
- No-go from the equivalence principle: All gravity probes (dynamics, lensing, photometry) agree on the same M_{true}; created mass would break this.

Q3.3a

A major conceptual issue concerns the origin and uniqueness of the relationship between the Mezzi scale factor $\zeta(r)$ and the baryonic mass of galaxies. This relationship is not theoretically derived but is first implicitly imposed in Section 4, where $\zeta(r)$ is used to rescale radii, areas, and ultimately the enclosed mass. From this point onward, all key results, mass amplification, deflection angles, the RAR mapping, and the rotation-curve reconstruction, depend critically on the assumed ζ–mass connection. However, the ζ–mass relation exhibits very large scatter in the sample, indicating that it is not uniquely determined by the data. The choice of the constant C=377, which sets the overall amplitude of $\zeta(r)$, appears to be selected to optimize agreement with the SPARC sample rather than derived from first principles.

A3.3a

The reviewer asserts that the ζ-mass relation "exhibits very large scatter in the sample" and is not uniquely determined. The paper states the exact opposite in Section 4.6, where a high correlation coefficient (specifically r = 0.96) is reported for the total mass ratio vs. the Mezzi scale factor, indicating that the scaling is "robust across three orders of magnitude in stellar mass". This assertion misrepresents the statistical results.

Furthermore, the reviewer's claim that C = 377 is merely "selected to optimize agreement" mischaracterizes both the SPARC dataset and the mathematics through which the ζ-M figures are generated.

Step 1 (Data Input): The observed enclosed baryonic mass profile $M_{obs}(<r_{obs})$ is fixed directly from SPARC photometry and kinematics using standard mass-to-light ratios (Eqs. 1-4). No dark matter is added, and no mass-to-light ratios are tuned.

Step 2 (Flow Field): The Painlevé-Gullstrand velocity field v(r) is computed strictly from this observed mass profile via Eq. 5.

Step 3 (Path Integral): The dimensionless kinematic accumulation $A(r)$ is calculated by integrating this velocity field along the line of sight from the galaxy's emission radius to its observed distance D (Eq. 7). This step is entirely determined by the galaxy's mass distribution and distance.

Step 4 : The kinematic compression $K(r) = \exp[A(r)]$ is evaluated (Eq. 8).

Step 5 : The universal compliance constant C is applied to convert the compression factor into the Mezzi scale factor: $\zeta(r) = 1/\exp[-C \cdot (K(r)-1)]$ (Eq. 12).

Step 6: The observed radial grid is transformed into the true radial grid using $r_{true} = r_{obs} / \zeta(r_{obs})$ (Eq. 13).

Step 7 : The true annular areas ΔA_{true} are calculated geometrically from the new r_{true} boundaries (Eq. 17). The true shell masses are then reconstructed by remapping the observed local density onto the correct, larger true coordinate areas : $\Delta M_{true,j} = \Sigma_{obs,j} \cdot \Delta A_{true,j}$ (Eq. 18).

Step 8: The true total mass M_{true} is calculated by summing these discrete shells (Eq. 20).

The ζ-mass scaling relation plotted in Figure 7 is the output of these 8-steps, not an input. Because $\zeta(r)$ is derived directly from the integral of the observed baryonic mass profile (Steps 2-5).

Q 3.3b

Because the ζ–mass relation is neither theoretically constrained nor uniquely determined by the observations, the framework currently lacks predictive power. The manuscript should explicitly justify how $\zeta(r)$ and the constant C are chosen, explain why this choice is preferred over other possible mappings, and demonstrate that the results do not depend sensitively on this tuning. Without such clarification, it is difficult to assess whether the Mezzi effect represents a physical mechanism or an empirical rescaling calibrated to match the data.

A 3.3b

The framework results in emergent scaling relations that arise from a single C. These predictions include:

1. Universal Surface Density Scaling Relation

$\Sigma_{\text{true}}/\Sigma_{\text{obs}} \propto (R_{\text{true}}/R_{\text{obs}})^{0.5}$ with $R^2 = 0.85$.

The exponent 0.5 is not imposed. It emerged.

2. Linear Mass Radius Scaling

$M_{\text{obs}}/M_{\text{true}} \propto r_{\text{obs}}/r_{\text{true}}$ with slope $0.95 \approx 1.0$ over three orders of magnitude.

If area scales as r^2, one might expect $M \propto r^2$. Near linear scaling emerged.

3. Rank Order Preservation of Central Surface Density

Correlation ~ 1.0 between $\Sigma_{0,\text{obs}}$ and $\Sigma_{0,\text{true}}$ across all mass regimes.

The framework could have scattered the hierarchy. Instead, the monotonicity of $\zeta(r)$ preserves rank order.

4. Freeman's Law

The observed near universality of central surface density ($\Sigma_0 \approx$ constant) in the true frame, with scatter in the observed frame interpreted as geometric distortion.

The framework predicts that the true central densities of massive galaxies are more uniform than observed, suggesting Freeman's Law reflects intrinsic physics rather than observational selection.

5. Total Mass Ratio vs. Mezzi Scale Factor

$M_{\text{true}}^{\text{tot}}/M_{\text{obs}}^{\text{tot}} \propto \zeta^{1.84}$ with $r = 0.967$.

The exponent 1.84 (not 2.0) emerged.

6. Deflection Angle Ratio vs. Mezzi Scale Factor

$\alpha_{\text{true}}/\alpha_{\text{obs}} \propto \zeta^{1.26}$ with $r = 0.836$.

The shallower exponent compared to mass scaling (1.26 vs. 1.84) emerged.

7. Counter Example: UGC02487

A galaxy with $M_{\text{ratio}} = 3.25$ but $\alpha_{\text{ratio}} = 1.09$ (nearly unity).

The framework predicts that extreme geometric scaling can reduce lensing relative to the mass recovery.

8. Cross Correlation: Deflection Angle vs. Rotation Velocity

$\alpha_{\text{max}} \propto v_{\text{obs}}^{2.19}$ with $R^2 = 0.95$.

This validates that the same M_{true} generates both lensing and dynamics.

9. Emergence of the RAR

Test 2 scatter 0.294 dex vs. observed RAR scatter 0.269 dex (ratio 1.09).

The RAR is not built into the model. It emerges from mapping true frame baryonic accelerations onto observed frame dynamical accelerations via $\zeta(r)$.

10. MOND like Behavior from Pure Coordinate Error

Test 3 scatter 0.248 dex against deep MOND relation $g_{\text{obs}} = \sqrt{g_{\text{bar}} a_0}$.

The framework predicts that evaluating true mass at compressed radius produces MOND like scaling without modifying gravity.

11. BTFR Slope Transformation

True frame slope 2.81 ± 0.16 (approaching Newtonian v^2); Empirical mapping slope 3.70 ± 0.58 (approaching observed v^4).

Neither slope is assumed. The steepening emerges algebraically from combining Newtonian dynamics, geometric mapping, and Freeman's Law.

12. BTFR Normalization Independence

β_{obs} correlates with ζ ($r = 0.53$, slope 2.09); β_{true} does not ($r = 0.01$, slope 0.04).

If the framework created mass, β_{true} would still correlate with ζ.

13. Radial Evolution of g_{ratio}/a_0

In outer regions, median $g_{\text{ratio}}/a_0 \approx 0.60$, below the MOND regime.

The framework predicts that the apparent "MOND transition" is radially dependent and geometric, not a universal acceleration threshold.

The emergence of these relations and consistent dynamics from one global value constitutes strong predictive power and demonstrates that C captures a genuine physical scale.

Q4.1a

A central concern arises in the treatment of gravitational deflection. The manuscript appears to conflate the role of the three-dimensional galactic scale with the quantity that actually governs light bending in general relativity. In the weak-field limit, the deflection angle depends on the projected mass enclosed within the impact parameter, not on the physical size of the galaxy. The impact parameter is simply the closest approach of the light ray, and enlarging the three-dimensional radius of the system does not alter this quantity. As long as the projected mass increases, the deflection angle must increase accordingly.

A4.1a

We clarify this point in Section 5.1, noting that the Mezzi framework does not modify standard GR lensing relation; it modifies the inputs to it. In the framework, both the impact parameter and the enclosed mass are transformed: $b_{\text{true}} = b_{\text{obs}}/\zeta(b_{\text{obs}})$ and $M_{\text{true}}(<b_{\text{true}})$ is the reconstructed true mass. The deflection angle in the true frame is:

$$\alpha_{\text{true}} = \frac{4G}{c^2} \frac{M_{\text{true}}(<b_{\text{true}})}{b_{\text{true}}}$$

Because $\zeta(r)$ is radially dependent, i.e. stronger at center, weaker in outer regions, the true mass is distributed out to larger radii than inferred in the observed frame, resulting in a lower true projected surface density. This is why $\alpha_{\text{true}}/\alpha_{\text{obs}} \propto \zeta^{-1.26}$ scales more shallowly than $M_{\text{true}}/M_{\text{obs}} \propto \zeta^{-1.84}$. The framework explicitly calculates and discusses this density lowering effect (Section 5.1).

Q4.1b

In the proposed framework, however, the authors argue that the enhancement in mass can be offset by an increase in the "true" galactic radius, leading in some cases to little or no change in the predicted deflection angle. This interpretation is difficult to reconcile with the standard GR definition of weak-lensing deflection, where the three-dimensional extent of the mass distribution plays no direct role. If the true mass is indeed several times larger than the observed mass, the corresponding lensing signal should be stronger, independent of the system's intrinsic spatial scale.

A4.1b

The reviewer implies that the proposed work scales the geometry uniformly, overlooking the radial dependence detailed in Section 4. The Mezzi scale factor $\zeta(r)$ is radially dependent, exhibiting stronger compression in the center and weaker effects in the outer regions, the larger true radial extent of the radius r_{true} indeed offsets the increase in the true enclosed mass M_{true}. Consequently, the framework naturally derives distinct, non linear scaling laws for dynamics versus lensing across the different galaxies. As shown in Figure 9 and Section 5.1, the total mass scales as $M_{true}/M_{obs} \propto \zeta^{-1.84}$, while the deflection angle scales more shallowly as $\alpha_{true}/\alpha_{obs} \propto \zeta^{-1.26}$. This shallower slope is the direct, unavoidable mathematical consequence of the mass being distributed across a larger physical area, effectively revealing a lower projected surface density that governs lensing efficiency. Again, this is derived from a universally fixed parameter, without per galaxy tuning.

The framework explicitly discusses scenarios where mass recovery is neutralized by the lower true surface density. We highlight the specific counter example of galaxy UGC02487, discussed in Section 5.1. Despite falling into the High Correction category with a substantial mass ratio of $M_{ratio} = 3.25$, it exhibits a deflection ratio of $\alpha_{ratio} = 1.09$ (nearly unity). This occurs precisely because the geometric correction reveals the mass is distributed over a much larger true radial extent, meaning the intrinsic surface density is lower, neutralizing the lensing increased lensing signal that would occur if the mass were concentrated at the observed radius. This is not a conflation of scales; it is the exact physical mechanism by which the framework remains consistent with weak field observations.

As we now emphasize in the revised Section 5.1, this differential scaling between dynamical mass ($\propto \zeta^{-1.84}$) and lensing deflection ($\propto \zeta^{-1.26}$) provides empirical consistency that the framework conserves mass and does not create it ad hoc. If the Mezzi effect were simply injecting mass independently of geometry, the deflection angle would scale linearly with the total mass recovery. The shallower scaling of the deflection angle supports that the true mass profile is physically spread over a proportionally larger volume, partially offsetting the gravitational pull of the recovered mass. This is the exact signature of correcting a radial misattribution, where the same baryons occupy a larger true volume than the observer infers.

Q4.1c
Given this discrepancy, the manuscript should clearly explain how the Mezzi framework modifies the GR lensing relation, why the usual dependence on projected mass no longer applies and how the proposed behavior remains consistent with existing weak-lensing observations. Without such clarification, it is challenging for the reader to understand why the new result should be preferred over the well-tested GR prediction.

A4.1c
The reviewer demands an explanation of how the Mezzi framework modifies the GR lensing relation. While it does not.

The manuscript does not claim that enlarging the 3D radius leaves the deflection angle unchanged if projected mass increases in a static geometry. Instead, it demonstrates that in the Mezzi framework, the process of recovering the true mass also involves mapping to a larger true radius in a way that lowers the projected surface density. Naturally resulting in lensing predictions that are consistent with the standard GR weak field approximation.

The physics of weak field lensing remains entirely standard; this framework addresses the geometric baseline of the mass distribution being lensed. The distinct scaling $\alpha_{\text{true}}/\alpha_{\text{obs}} \propto \zeta^{-1.26}$ vs. $M_{\text{true}}/M_{\text{obs}} \propto \zeta^{-1.84}$ provides a falsifiable prediction.

Furthermore, the cross-analysis in Section 5.2 validates the physical consistency of this interpretation. We now explicitly note in the manuscript that if M_{true} represented "created" mass rather than remapped baryons, the deflection angle $\alpha_{\text{true}} \propto M_{\text{true}}/r_{\text{true}}$ would bear no necessary relationship to the observed rotation velocity v_{obs}, which is measured independently from Doppler shifts. The fact that $\alpha_{\text{true}} \propto v_{\text{obs}}^{2.19}$ with $R^2 = 0.95$ confirms that M_{true} is the physically correct baryonic mass distribution, the same matter generating both the lensing potential and the rotational dynamics. This near-unity correlation would be a remarkable coincidence if mass were being invented; it is the expected consequence if the same baryons are simply distributed over their true spatial extent.

Q5.1a

A major concern arises from the interpretation of Figures 20-21. While the authors emphasize that the β-ζ relation in the true frame exhibits an approximately zero slope, the vertical scatter of the normalization offset is extremely large, spanning roughly 1–3 dex. This level of dispersion is physically implausible for the baryonic Tully–Fisher relation. Observationally, the BTFR is one of the tightest empirical relations in galaxy dynamics, with a typical intrinsic scatter of only 0.1–0.2 dex (Lelli et al. 2019 MNRAS). Even the most extreme outliers rarely exceed 0.3 dex.

A5.1a

There appears to be a confusion in between the BTFR normalization offset (Figures 20-21) and the intrinsic scatter of the BTFR (Figure 19). The scatter is 0.31 dex, not 1-3 dex. On the other hand, the scatter around the BTFR offset slope line, as laid out in in Figures 20 and 21, across the three mass regimes is 0.27 dex (High mass), 0.25 dex (Intermediate mass), and 0.49 dex (Low mass).

The paper's claim in this test is that the systematic trend of β with ζ disappears in the true frame. Figure 20 shows β_{obs} correlates strongly with ζ (slope 2.09, r=0.53); Figure 21 shows β_{true} has slope 0.04 (r=0.01). The text in Section 8.2 explicitly states " confirming that the normalization offset is no longer dependent on geometric compression. The scatter of the relation remains consistent across mass bins...".

Q5.1b

A β offset varying by 1–3 dex corresponds to mass discrepancies of one to three orders of magnitude. Such a wide spread indicates that the reconstructed "true" masses are highly unstable and that the \zeta–mass mapping lacks uniqueness or physical constraint. In this context, the disappearance of the slope in Figures 20-21 cannot be taken as evidence that the BTFR normalization has been "restored." Instead, the enormous scatter demonstrates that the model fails to reproduce the defining tightness of the BTFR, which is essential to any credible explanation of its physical origin. In short, the conclusion drawn from Figures 20-21 is not justified. A zero slope with a scatter of several dex does not support the proposed geometric interpretation; rather, it highlights a fundamental inconsistency in the mass reconstruction and the assumed \zeta(r) scaling.

A5.1b
With the correct scatter established, we strongly disagree with the reviewer's underlying premise. On the contrary, the transition from the true frame to the observer frame is a mathematically consistent and physically meaningful consequence of the proposed framework. The geometric distortion \zeta scales systematically with the depth of the gravitational potential. This projection transforms the intrinsic true frame BTFR slope of ~2.8 into the observed BTFR slope of ~3.7, closely matching the empirical value of ~4. Massive galaxies, which would otherwise deviate significantly in the true frame, are compressed more severely, pulling their observed masses and velocities onto the tight, steep observed BTFR line.

This shift in slope is profoundly meaningful: it demonstrates that the observed BTFR slope of ~ 4 is not a fundamental law of nature, but an emergent artifact of geometric projection.

We have revised Sections 8.1, 8.2, and 8.3 to make the conservation of baryonic mass in this derivation explicit:

1. The Geometric Derivation: The steepening from slope \sim 2 to \sim 4 is a mathematical consequence of three premises: (i) Newtonian gravity in true coordinates, (ii) constant observed surface density (guaranteed approximately by SBI), and (iii) linear mass-radius geometric scaling. The v^4 scaling emerges from the algebraic elimination of r_{\text{obs}}. At no stage is mass created or destroyed; the exponent 4 is the predicted consequence of mass conservation under geometric compression.

2. The Normalization Offset: The vanishing of the \beta-\zeta correlation in the true frame (r = 0.01, slope 0.04) is the strongest empirical evidence against mass creation. If the Mezzi effect invented mass, \beta_{\text{true}} would correlate with \zeta. Instead, \beta_{\text{true}} becomes independent of geometric compression, indicating M_{\text{true}} is the intrinsic physical mass. The residual scatter reflects genuine astrophysical variance rather than geometric uncertainty, the hallmark of a measurement calibration, not a mass-generation mechanism.

Q6.0
The implications in Section 9 are not internally consistent with the assumptions of the Mezzi framework.

A6.0
The implications in Section 9 are forward looking predictions of the theoretical framework, they identify how the core axioms, vacuum compliance, path dependent accumulation, and geometric distortion, should manifest when the mathematical implementation is extended to realistic geometries. It is ogically inconsistent to attempt to contradict the theoretical framework itself based on the known, deliberate simplifications. Such predictions guide future model development and observational tests.

Q6.1
The model assumes a spherically symmetric vacuum flow, yet the inclination-dependent dark matter fraction requires a strongly anisotropic, disk-plane–dominated compression. These two assumptions cannot both be true.

A6.1
The reviewer mistakenly treats a predicted observational signature of the proposed theoretical framework as an internal logical contradiction based on a known simplification of the current mathematical model.
The prediction that edges on galaxies exhibit higher inferred dark matter fractions than face on systems follows directly from the physical logic of the framework. The mezzi effect acts more strongly in the radial plane of the disk, so line of sight integration through the plane of maximum compression effect should amplify the apparent mass discrepancy relative to integration along the axis of minimal compression. This is the qualitative physical consequence of the theory.

Q6.2
Earlier sections state that \zeta(r) is determined solely by the galaxy's own baryonic potential. Section 9.2 then claims \zeta(r) is strongly modified by the host cluster potential. This contradicts the model's own definition of \zeta(r).

A6.2
As addressed in the previous point, the isolation assumption is a boundary condition for the path integral (Section 3.5). It is not a fundamental axiom of the framework that galaxies must be physically isolated for the Mezzi effect to exist.
In a cluster environment, overlapping potentials would simply alter the boundary conditions and the integration path of the accumulation integral. The framework predicts that non isolated environments will exhibit modified distortion profiles. This is a direct consequence of the theory to be tested against observation.

Q6.3
The model states that higher concentration produces stronger compression (smaller \zeta). Section 9.4 claims compact globular clusters should have weak compression (\zeta \approx 1). This reverses the model's own scaling rule.

A6.3

This assumption is incorrect because compactness alone does not drive the Mezzi effect. It is stated in the paper's Eq 5 and Eq 7, the velocity flow $v(r)$ and the accumulation integral $A(r)$ are driven by $M(r)$. The Mezzi scale factor $\zeta(r)$ depends on the “gravitational potential depth”, i.e. mass and enclosed radius, not merely on “compactness”. Globular clusters have very low total mass compared to galaxies. Even if they are highly concentrated, their shallow total gravitational potential results in a negligible flow velocity $v(r)$, leading to $A(r) \approx 0$ and $\zeta \approx 1$, overlooking the dominant variable of total enclosed mass defined in Eqs. 5 and 7.

Q6.4

The mathematical form of $\zeta(r)$ steepens the inner mass profile (enhanced compression). Section 9.5 claims it naturally produces cores. The predicted geometry actually produces cusps, not cores.

A6.4

The claim that the mathematical form of $\zeta(r)$ 'produces cusps, not cores' inverts the direction of the Mezzi transformation. The framework suggests the exact opposite.

The Mezzi framework does not start with a true profile and compresses it forward into an observed cusp. It starts with observed photometry and reconstructs the true geometry via $r_{\text{true}} = r_{\text{obs}}/\zeta(r_{\text{obs}})$. Because $\zeta(r)$ reaches its minimum at the galactic center, the inverse amplification factor $1/\zeta(r)$ is greatest in the innermost region . This means the central shells are mapped to larger true areas more than the outer shells during reconstruction.

Since ζ increases with radius, the area ratio $R_j = \Delta A_{\text{true},j}/\Delta A_{\text{obs},j}$ is maximal for inner shells. The same observed mass element, when mapped back to its correct, disproportionately larger true area, reveals a true surface density that is inherently lower at the center than the observer infers. :

$$\Sigma_{\text{true},j} = \frac{\Sigma_{\text{obs},j}}{R_j}$$

This is the direct output of the reconstruction. Figure 2 (right panel) empirically confirms this: for high mass galaxies, $\Sigma_{0,\text{true}}/\Sigma_{0,\text{obs}} \approx 0.6$, with the ratio decreasing systematically as mass increases. The true central density is lower than the observed value.

The physical consequence is that the true cumulative mass profile grows more gradually in the inner regions. The mass that is perceived to be concentrated in a small observed central volume is, in the true frame, actually distributed across a larger physical volume. The resulting true density slope flattens. The core is not an assumption imposed on the true frame; it is an emergent mathematical property of inverting the observed compression.

Q6.5

The model asserts \zeta(r) only rescales coordinates without altering physical dynamics. Section 9.6 requires \zeta(r) to change the true mass distribution enough to fit Einstein radii. A pure coordinate transformation cannot generate additional lensing mass.

A6.5

As established in the response to 3.1a, the Mezzi effect suggest that our observation integrates the correct local density over a compressed volume, systematically undercounting the existing mass. The Mezzi scale factor \zeta(r) corrects this undercounting by correcting the integration volume to the true geometry. Because this rescaling is radially dependent, it remaps the mass over a larger volume, which lowers the projected surface density.

Dear Author,

Thank you for submitting your manuscript to Physics of the Dark Universe.

We invited an independent expert to review your appeal and revised version. Please see attached Report of Referee-2. In view of this new Report, we regret to inform you that the Referee-2 recommended against publishing your manuscript.

Reviewer 2 - Revision 2, July 12

Basically, I agree with almost all points of first referee who did very careful evaluation of this work. It contains basic errors and should not be published. Moreover, i do not see any novel points or even idications towards to novel points. The fact that observational data depend somehow from frame/observer is well-known. The corresponding observer independent or frame independent consideration is often necessary. Without repeating detail of this work I suggest to reject it.